TASK 1 — Agentic AI System for Multi-Step Tasks

Objective
Build a system that can accept a complex user request, decompose it into discrete steps, assign each step to a specialized agent, stream partial results, and handle failures gracefully.

Requirements
Your system must demonstrate the following capabilities:
• Accept a complex, multi-part task as input
• Decompose the task into discrete, ordered steps
• Use multiple specialized agents (e.g., Retriever, Analyzer, Writer)
• Execute using async or pipeline-based architecture
• Stream partial outputs to the user
• Include meaningful failure handling logic
• Implement manual batching logic (no black-box abstraction)

Deliverables
• GitHub repository with clean, well-commented commits
• System design document covering architecture and data flow
• 3–5 minute explanation video (must demonstrate the system running and show at least one failure case)
• Post-mortem document including:
• One scaling issue you encountered or anticipate
• One design change you would make in hindsight
• Two explicit trade-offs and your reasoning

Constraint: Do NOT rely on black-box agent frameworks without explicit justification. Show that you understand what is happening under the hood.

What We Evaluate
✓ Execution ability
✓ System structuring
✓ Practical problem-solving
✓ Basic ownership